



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Divya Konar
Roll No. :	20
Experiment No.:	4
Title of the Experiment:	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets
Date of Performance:	12/08/26
Date of Submission:	19/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi P.N.**

Signature :

Date :



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Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:
 - a. One-sample t-test
 - b. Two-sample t-test
 - c. Chi-square test



7. Calculate the test statistic and p-value.
8. Compare the p-value with the selected significance level.
9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ .

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with a particular characteristic.

Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$



The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p - value < \alpha$, reject H_0 .
- If $p - value \geq \alpha$, fail to reject H_0 .

t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test



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The chi-square test can be used to test the association between categorical variables.

For example:

Diabetes Outcome and Gender

where applicable.

Output:

```
OUTPUT ... Filter (e.g. text, excludeText, t...) Code
[Running] python -u "e:\MYDATA\STATS\exp4.py"
=====
STATISTICAL ANALYSIS OF PIMA INDIANS DIABETES DATASET
=====

1. DATASET LOADED SUCCESSFULLY
-----
Number of rows: 768
Number of columns: 9

First 5 rows:
| Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72  ...              0.627    50        1
1            1       85             66  ...              0.351    31        0
2            8      183             64  ...              0.672    32        1
3            1       89             66  ...              0.167    21        0
4            0      137             40  ...              2.288    33        1

[5 rows x 9 columns]

2. DATASET INFORMATION
-----

Column names:
['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI',
'DiabetesPedigreeFunction', 'Age', 'Outcome']

Data types:
Pregnancies          int64
Glucose              int64
BloodPressure        int64
SkinThickness        int64
- ...
```



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```
OUTPUT ... Filter (e.g. text, lexcludeText, t... Code
=====
SkinThickness      int64
Insulin            int64
BMI                float64
DiabetesPedigreeFunction float64
Age               int64
Outcome            int64
dtype: object

Missing values:
Pregnancies      0
Glucose          0
BloodPressure    0
SkinThickness    0
Insulin          0
BMI              0
DiabetesPedigreeFunction 0
Age              0
Outcome          0
dtype: int64
```

3. SELECTED VARIABLES

Numerical variables: Glucose, BMI, Age
Categorical variable: Outcome
Outcome: 0 = Non-diabetic, 1 = Diabetic

4. GROUPING THE DATA

Non-diabetic group: 500
Diabetic group: 268

```
OUTPUT ... Filter (e.g. text, lexcludeText, t... Code
=====
PART A: POINT ESTIMATES
=====

5. SAMPLE MEANS
-----
Mean Glucose = 120.89
Mean BMI      = 31.99
Mean Age      = 33.24

6. SAMPLE PROPORTION
-----
Number of diabetic individuals: 268
Total individuals: 768
Sample proportion of diabetic individuals = 0.3490
Sample proportion (%) = 34.90%

=====
PART B: CONFIDENCE INTERVALS
=====

7. 95% CONFIDENCE INTERVAL FOR MEAN GLUCOSE
-----
Sample mean = 120.89
Sample standard deviation = 31.97
Sample size = 768
Margin of error = 2.26
95% Confidence Interval = (118.63, 123.16)

Interpretation: We are 95% confident that the population mean glucose level lies
between 118.63 and 123.16.
```



OUTPUT ...

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Code

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8. 95% CONFIDENCE INTERVAL FOR DIABETES PROPORTION

Sample proportion = 0.3490

Margin of error = 0.0337

95% Confidence Interval = (0.3152, 0.3827)

In percentage terms: (31.52%, 38.27%)

Interpretation: We are 95% confident that the true population proportion of diabetic individuals lies between 31.52% and 38.27%.

PART C: ONE-SAMPLE T-TEST

9. ONE-SAMPLE T-TEST

Research Question:

Is the population mean glucose level significantly different from 120?

Null Hypothesis (H_0):

The population mean glucose level is equal to 120.

Alternative Hypothesis (H_1):

The population mean glucose level is not equal to 120.

Significance level (α) = 0.05

Sample mean = 120.89

Test statistic (t) = 0.7754

p-value = 0.438372

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Decision: Fail to reject H_0 .

Interpretation: There is insufficient statistical evidence to conclude that the population mean glucose level is significantly different from 120.

PART D: TWO-SAMPLE T-TEST

10. TWO-SAMPLE T-TEST

Research Question:

Is there a significant difference in mean glucose levels between diabetic and non-diabetic individuals?

Null Hypothesis (H_0):

There is no significant difference in mean glucose levels between the two groups.

Alternative Hypothesis (H_1):

There is a significant difference in mean glucose levels between the two groups.

Significance level (α) = 0.05

Mean glucose - Diabetic group = 141.26

Mean glucose - Non-diabetic group = 109.98

Test statistic (t) = 13.7515

p-value = 0.000000

Decision: Reject H_0 .

Interpretation: There is a statistically significant difference in mean glucose levels between diabetic and non-diabetic individuals.



OUTPUT ...

Filter (e.g. text, !excludeText, t...

Code

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PART E: CHI-SQUARE TEST

11. CHI-SQUARE TEST

Contingency Table:

	Non-Diabetic	Diabetic
AgeGroup		
Young (<=30)	327	90
Middle (31-50)	130	140
Older (>50)	43	38

Research Question:

Is there a significant association between age group and diabetes status?

Null Hypothesis (H0):

Age group and diabetes status are independent.

Alternative Hypothesis (H1):

Age group and diabetes status are associated.

Significance level (alpha) = 0.05

Chi-square test statistic = 71.8502

Degrees of freedom = 2

p-value = 0.000000

OUTPUT ...

Filter (e.g. text, !excludeText, t...

Code

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Expected Frequencies:

	Non-Diabetic	Diabetic
AgeGroup		
Young (<=30)	271.48	145.52
Middle (31-50)	175.78	94.22
Older (>50)	52.73	28.27

Decision: Reject H0.

Interpretation: There is a statistically significant association between age group and diabetes status.

FINAL SUMMARY OF RESULTS

Point Estimates:

Mean Glucose = 120.89

Mean BMI = 31.99

Mean Age = 33.24

Diabetes Proportion = 0.3490

Confidence Intervals:

95% CI for Mean Glucose = (118.63, 123.16)

95% CI for Diabetes Proportion = (0.3152, 0.3827)



Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

ANS: Population Parameter Estimated:

The population mean glucose level was estimated using the sample mean. The estimated mean glucose level was approximately 120.89.

2. What confidence interval was obtained, and how should it be interpreted?

ANS: A 95% confidence interval for the population mean glucose was obtained as approximately (116.88, 124.90). This means we are 95% confident that the true population mean glucose level lies within this range.

3. What hypotheses were formulated for the selected statistical test?

ANS: Null Hypothesis (H_0): The population mean glucose level is equal to 120.

Alternative Hypothesis (H_1): The population mean glucose level is not equal to 120.

A one-sample t-test was used to test these hypotheses.

4. Based on the p-value, what statistical decision was made?

ANS: The obtained p-value was greater than 0.05. Therefore, we failed to reject H_0 . This indicates that there is insufficient statistical evidence to conclude that the population mean glucose level is significantly different from 120.



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